EXPERT, STEP 23

A workflow with a stop rule

Keep fixing until the test passes. Who decides when to stop?

```
# python3 test_changes.py, before the fix
FAIL: changes.py does not print the three expected lines.
Expected:
  2. route  NL-BE    77 ->  101  +24
Got:
  2. client Epsilon Garden (example)     9 ->   20  +11
```

THE SETUP

A small script with one planted bug, a test, and two Never lines in CLAUDE.md.

THE TEST

The same ask, plain and as a workflow, 5 sessions each. Then a test that cannot pass.

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The plain ask fixed it too

Both asks fixed the bug. The plain ask used far fewer tokens and no more questions.

```
# plain ask, the edit on line 52
-    changes.sort(key=lambda c: (-abs(c[5]), c[1], c[0]))
+    changes.sort(key=lambda c: (-abs(c[4]), c[1], c[0]))
# then the start of Claude's answer
The test passed on the second run, after one fix.
```

PLAIN ASK

Test passed in 5 of 5. 2 or 3 permission questions. About 118,000 to 162,000 tokens, most of them cache reads.

WORKFLOW ASK

Test passed in 5 of 5. 3 or 4 permission questions, plus the workflow question. About 306,000 to 361,000 tokens, most of them cache reads.

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Read the plan before it runs

In accept edits mode, a workflow asks before it runs, and it can show you the script.

```
# the workflow script, lines 31 to 33
if (r.failing_count < best) { best = r.failing_count;
noProgress = 0 } else noProgress++
}
if (noProgress >= 2) return { status: 'stalled', rounds:
round, history }
```

THE QUESTION

Run a dynamic workflow? For a new script, option 2 shows the raw script.

THE STOP RULE

Two lines in the script: no progress counts up, and 2 rounds without progress stop the loop.

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While it runs

The workflow runs in the background. /workflows shows each round.

```
# /workflows, after the run
fix-changes-loop
1/1 agent · 13s · done
✔ round 1       Opus 5.5 · 29.9k tok   13s
```

THE VIEW

One phase, one agent per round, with its time.

THE QUESTIONS

The workflow agent asks before most of its shell commands: twice in each one-round run here.

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The stop rule, on a test that cannot pass

We made the test expect -25, while 48 minus 71 is -23.

```
# step 10 of the kit, from Claude's final report
The workflow has a bug: it ran 8 rounds instead of
stopping after 2 without progress.
```

PLAIN ASK

One fix, then Claude stopped by its own judgement. 2 of 2.

WORKFLOW

3 of 4 runs stopped on the no-progress rule after 2 or 3 rounds. 2 of them counted the real fix as no progress. 1 ran to its cap of 8 rounds. Claude changed neither the test nor the CSV.

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Save it, run it again

Save the script from /workflows with s. In a new session it is a slash command.

```
# after s in /workflows, then Enter
Dynamic workflow saved to
.../.claude/workflows/fix-changes-loop.js. Invoke as
/fix-changes-loop ...
```

BEFORE YOU SAY YES

- [ ] Find the loop.
- [ ] Find the stop rule.
- [ ] See how it measures progress.
- [ ] Look for a round cap.
- [ ] See which files it tells the agent to leave alone.

TRY IT

The test folder, our saved script and the test that cannot pass are in the AI Starter Kit, Expert, step 23.